PART I: FILL IN THE BLANK SPACE ITEMS

Direction: Fill in the blank space with the most simplified answers [9%, 1.5pt each blank space]

1. If the temperature of a cube made of copper with initial volume 8 m^3 changes from 10 °C to 20 °C, its change in volume is [redacted] m^3 (Use: $\alpha = 1.7\times10^{-5}K^{-1}$)
2. In a car lift used in a service station, compressed air exerts a force on a small piston that has a circular cross section and a radius of 2 cm. This pressure is transmitted by a liquid to a piston that has a radius of 4cm and 1.2×10^4 N is used to lift a car weight. The force that must be exerted is [redacted] N
3. Two resistors connected in series have an equivalent resistance of 16 Ω. When they are connected in parallel, their equivalent resistance is 3 Ω. The resistances of the two resistors are [redacted] and [redacted]
4. Four point charges are located at the corners of a square. If each charge has magnitude 6 μC and the square has sides of length 4 cm, the magnitude of the electric field at the center of the square is [redacted] N/C
5. A proton (q = 1.6×10^{-19} C) moves with a velocity of **v** = (2**i** - 4**j** + **k**) m/s in a region in which the magnetic field **B** = (**i** + 2**j** - 3**k**)T. The magnetic force experienced by the proton is [redacted] $\times10^{-19}$ N
6. A 0.5-mol sample of an ideal gas is kept at a constant temperature of 300K expands isothermally from 2L to 6 L. The energy transferred by heat between the gas and the surrounding during this process will be [redacted] J. (Take R = $\frac{8.31 J}{mol.K}$)

PART II: SHORT ANSWER ITEMS

Direction: Give short answers for the following questions only on the space provided [6%, 1pt each]

1. List two heat transfer mechanisms and explain their differences

There are 4 types of Heat transfer mechanism
- Conduction
- Convection
- radition
- direct burning

Conduction :-> It occur when we transfer heat by direct contact
-> with in a medium

Convection:- It occur, with in a medium like Conduction but, there is no direct contact.

Radiation :- It occur, ~~there~~ there is no contact b/n the object a heat source and also there is no medium

direct burning -> It occur by direct burning of the object

2. A raw egg dropped to a cement floor breaks upon impact, but when the raw egg is dropped on to a thick foam mattress, it rebounds without breaking, why is this possible?

I think may be because of may be the egg dropped in the floor on the area that the mass is equally distributed like this , but if it dropped , it will be come break. law of pressure $P = \frac{F}{A}$, the force equally distributed on hole area

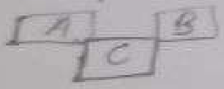
3. State and explain the following laws:

i) Kepler's third law of planetary motion (law of Harmony)

State that "square of orbital period of a planet directly proportional to the average distance from the center of the sun to the planet" $T^2 \propto R^3$, $\frac{T^2}{R^3}$ = Constant.

ii) The Zeroth law of thermodynamics

"If object A and B are separately thermal equilibrium with C object A and B are also thermal equilibrium each other."



iii) Lenz's law

✓ Work on Faraday's law of electromagnetic induction

✓ It say that the electromotive force occur in the induct process opposite in direction with rate of change of flux.

$Emf = -\frac{N\phi}{\Delta t}$ → the Negative sign indicate opposition b/n Emf & $\frac{\Delta\phi}{\Delta t}$

✓ Indicate direction of Induce emf

iv) Faraday's Law of electromagnetic induction

State that the Emf is directly proportional to it rate of change of flux.

$Emf \propto \frac{\phi}{\Delta t}$, $Emf = -\frac{N\phi}{\Delta t}$ → N = No of turn, ϕ = magnetic flux, Δt = time

– indicated by Lenz's law

✓ State only the magnitude of Induced Emf

PART III: MULTIPLE CHOICE ITEMS

Direction: choose the correct answer and write the letter of your choice only on the space provided [10%, 2points each]

[redacted] 1. Four objects are situated on xy-plane as follows: a 1 kg object at the origin, 2kg object at +1cm on the x-axis, a 3kg object at +2cm on the y-axis, and a 4kg object at (+1cm, -2cm). Where is the center of mass of these objects?

A. (1,0)cm B. (0.6, 1.4) cm C. (0.6, -0.2) cm D. (0.6, 0.2) cm

[redacted] 2. A tourist guide wanted to make tea in a cold morning at the top of Ras Dashen. Instead of water he got an ice at -10°C. He knows that water boils at that attitude at 90°C and freezes at 0°C. How much electrical energy should he use to get boiling water of 0.10kg to make tea?
(take L_f =334kJ/kg, c_{ice} =2.1kJ/kg°C , c_{water} =4.2kJ/kg°C)

A. 76.82kJ B. 170.9kJ C. 73.3J D. 73.3kJ

[redacted] 3. An object hangs from a spring balance. The balance indicates 30N in air and 20N when the object is totally submerged in water with density 1000kg/m³. What is the volume of the object? (take g = 10N/kg)

A. $10m^3$ B. $10^{-1}m^3$ C. $10^{-6}m^3$ D. $10^{-3}m^3$

[redacted] 4. Body A of mass 4m moving with speed u collides with another body B of mass 2m, at rest. The collision is head on and elastic in nature. After the collision the fraction of energy lost by the colliding body A is

A. 5/9 B. 1/9 C. 8/9 D. 4/9

[redacted] 5. A block of mass **m** is released from rest at point P and slides along the frictionless track shown. At point **Q**, its speed is:

A. $2g\sqrt{h_1 - h_2}$ B. $2g(h_1 - h_2)$ C. $\frac{h_1 - h_2}{2g}$ D. $\sqrt{2g(h_1 - h_2)}$

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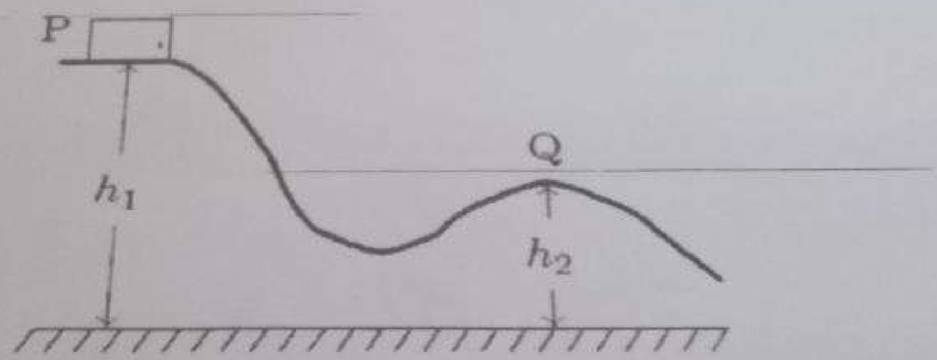


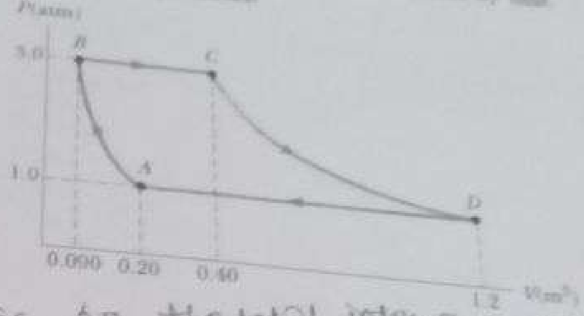
2. A sample of an ideal gas goes through the process shown in the Fig below. From A to B, the process is adiabatic; from B to C, it is isobaric with 100 kJ of energy entering the system by heat. From C to D, the process is isothermal; from D to A, it is isobaric with 150 kJ of leaving the system by heat. Determine the difference/change in internal energies for the three processes.

(Take 1atm=101.3KPa)

a) ΔU_{CD} [2pts]

b) ΔU_{AD} [2pts]

c) ΔU_{AB} [2pts]



A to B adiabatic

B to C Isobaric $Q = 100\,kJ$

C to D Isothermal

D to A Isobaric $Q = -150\,kJ$

the hole process is cyclic so the total internal energy

$E_{Total} = E_{AB} + E_{BC} + E_{CD} + E_{DA} = 0$

a. ΔU_{CD}, the the process is isothermal so

$\Delta U_{CD} = 0$

b. $\Delta U_{AD} = Q + W$ → $W = -P\Delta V$, we have the area under A & D

$= -(1\,atm)(0.2 - (1.2))\,m^3$

$= -(101.3\,KPa)(-1)\,m^3$

$= 101{,}300\,J$ it is positive

$= -150{,}000\,J + 101{,}300\,J$

$= -48{,}700\,J$

ΔU_{AB}, from the above equation

$E_{Total} = E_{AB} + E_{BC} + \cancel{E_{CD}} + E_{DA} = 0$

$U_{AB} = -U_{BC} + -U_{AD}$ → $\Delta W = -P\Delta V$

$= -(\Delta Q + \Delta W) + -(-48{,}700)\,J$

$= -(100\,kJ + (-94.2\,kJ)) + 48{,}700\,J$

$= -(5.8\,kJ) + (48.7\,kJ)$

$= 42.2\,kJ$

$= -(P_1)(V_2 - V_1)$

$= -(3\,atm)(0.4) - (0.09)$

$= -(303.9\,KPa)(0.31)\,m^3$

$= -94.2\,kJ$

PART IV: WORKOUT ITEMS

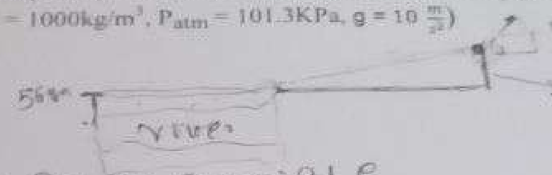
Direction: Solve the following problems by showing all the necessary steps briefly only on the space provided [25%]

1. Through a pipe 15.0 cm in diameter, water is pumped from the River up to a Village, located on the rim of the canyon. The river is at an elevation of 564 m, and the village is at an elevation of 2096 m.

a) What is the minimum pressure at which the water must be pumped if it is to arrive at the village? **[3pts]**

b) If 4500 m^3 are pumped per day, what is the speed of the water in the pipe? **[1pt]**

(Use density of water = 1000kg/m^3, P_{atm} = 101.3KPa, g = 10 $\frac{m}{s^2}$)



$h = 2096 - 564$

$h = 1532m$

a we use the pascal principle

$P = \rho g h \quad \rho = 1000 kg/m^3$

$= 1000 kg/m^3 \times 10 m/s^2 \times 1532 m$

$= 15{,}320{,}000 pa$

$P_T = P_{atm} + P_{guage}$

$= 101.3 kpa + 15320 kpa$

$= 15421.3 pa$

b, $V = 4500 m^3$

we use Bernoulli's principle

$P_0 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_0 + \frac{1}{2}\rho v_2^2 + \rho g h_2 \Rightarrow P = P$ is equal

$V_1 = 0 m/s$ B/c at rest

$\frac{1}{2}\rho v_2^2 = \rho g h_1 - \rho g h_2$

$\frac{1}{2} V_2^2 = g h_1 - g h_2$

$V_2^2 = \sqrt{2g(h_1 - h_2)}$

$A_1 v_1 = A_2 v_2$

$AV = AV$

(b) $\rho = \frac{m}{V} \Rightarrow m = \rho V = 1000 kg/m^3 \times 4500 m^3$

$m = 4500000 kg$ of water moved

$A v_1 = A_2 v_2$

Eight identical charges, each having a charge of $q = ?\ \mu C$, are fixed at the corners of a cube of side ... as shown in the Figure below.
a) Find the electric field at point P that lies a distance $z = 8$ m above the center of the cube, along the line perpendicular to the cube and passing through the center of the cube. [5 pts]
b) The net electrostatic force on a charge $Q = 2\mu C$ which is placed at point P [5pts]
c) The net electric potential due to the charges at point P [2 pts]

$E_r = E_{q1} + E_{q2} + E_{q3} + E_{q4} + E_{q5} + E_{q6} + E_{q7} + E_{q8}$

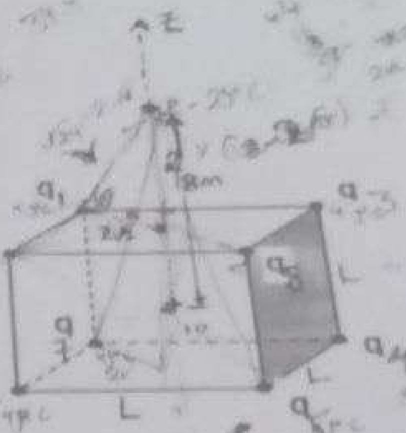
$E_1 = \frac{kq_1}{r_1^2} = \frac{9\times10^9 \times 4\times10^{-6} C}{(2\sqrt{6} m)^2} = \frac{36\times10^3}{24 m^2} = 1.5\times10^3 C$

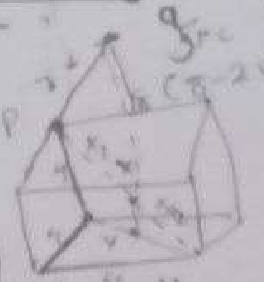
$E_{1x} + E_{1y} = E_{1x} = 1.5\times10^3 (\sin\theta)$, $E_{1y} = 1.5\times10^3(\cos\theta)$

$E_2 = \frac{kq}{r_2^2} = 1.5\times10^3$ N/C

E_3 and E_4 also the same with E_1, E_2

error the answer is in the back





(B) $F_{q_1 Q} = \frac{9\times10^9 \times 2\times10^{-6} \times 4\times10^{-6}}{42 m^2} = 1.7\times10^{-3} N$ $\theta = 59.5$

$F_x = (1.7\times10^{-3})N(\sin 59.5) = 1.4\times10^{-3} N$

$F_y = (1.7\times10^{-3})N(\cos 59.5) = 8.6\times10^{-4} N$

the magnitude is the same for q_1, q_2, q_3, q_4

$F_{q_2 Q} = (1.4\times10^{-3} i + 8.6\times10^{-4} j) N$

$F_{q_3 Q} = 1.4\times10^{-3}(-i) + 8.6\times10^{-4} j$ N

$F_{q_4 Q} = 1.4\times10^{-3}(-j) + 8.6\times10^{-4} j$ N

for $q_5 - q_8$

$F_{q_5 Q} = \frac{9\times10^9 \times 2\times10^{-6} \times 4\times10^{-6}}{(\sqrt{108} m)^2} = 6.67\times10^{-4} N$ $\theta = 74.2$

$F_x = 6.67\times10^{-4} N(\sin(74.2)) = 6.4\times10^{-4} N$

$F_y = 6.67\times10^{-4} N(\cos 74.2) = 1.8\times10^{-4} N$

the magnitude is the same

~~$F_{q_6 Q} = (6.4\times10^{-4} i + 1.8\times10^{-4}(i)) N$~~

$F_{q_7 Q} = (6.4\times10^{-4}(-j) + (1.8\times10^{-4}(j)) N$

$F_{q_8 Q} = (6.4\times10^{-4}(-j) + 1.8\times10^{-4}(j)) N$ [7]

$F_x = 0 N$

$F_y = (4(8.6\times10^{-4}) + 4(1.8\times10^{-4})) j N = 4.16\times10^{-3} j N$

error

first we find y

$y^2 = 4^2 + 4^2$

$y^2 = 32 \Rightarrow y = 4\sqrt{2}$

$x^2 = y^2 + 4^2$

$= (4\sqrt{2})^2 + 4^2$

$x^2 = 16\times2 + 4^2$

$x^2 = 32 + 16 = 48$

$x = 4\sqrt{3}$

to find r^2 for q_1

$r^2 = (2\sqrt{2})^2 - (8-...$

$r^2 = 8 - (0.43)$

$r^2 = 8 - 0.18$

$r^2 = \sqrt{7.8} = r =$

Back